Tristan Morton

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Education

James Madison University – B.B.A. in Computer Information Systems **GPA:** 3.2 Expected Dec 2027
Northern Virginia Community College – Dean's List, Scholar Honoree **GPA:** 3.9 Aug 2023 – May 2024

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, HTML/CSS
Technologies: PostgreSQL, pandas, scikit-learn, XGBoost, React, Next.js, Tailwind CSS, Streamlit, Docker, Git, GitHub Actions, pytest, MCP, Vercel, Microsoft Excel

Professional Experience

Software Engineering & AI Systems Intern, Formwork – Remote Sep 2026 – Present
JavaScript, TypeScript, Java, HTML, SQL, MCP, Linear

- Filed and worked **100+ QA tickets** in Linear covering bugs and feature gaps in Matter, a construction project management platform shipping **~3 releases per week** to a live customer.
- Helped integrate a **custom AI agent** into Matter over **MCP (Model Context Protocol)**: defined its workflows, connected APIs, wrote its prompts, and ran QA and accuracy reviews before it went live.
- Proposed and helped implement a **search filtering feature** that streamlined how users find records in the platform.
- Traced the platform's **database migration flow** end to end using **AI-assisted code analysis** with Claude, verifying each AI-generated claim against the source code.

Projects

Tripwire: Situational Awareness Dashboard | *Python, PostgreSQL, scikit-learn, Streamlit, Docker* Oct 2026 – Present

- Built an open-source dashboard that ingests the GDELT global news-event feed every **15 minutes**, storing **~800,000** conflict and unrest events in PostgreSQL with idempotent loads and failure handling.
- Designed spike detection using a **median/MAD robust z-score** against a **28-day baseline**; hand-audited **50 alerts** at **61% precision** (95% CI 47-74%) and rejected 3 tuning fixes that failed on a held-out set.
- Trained a logistic regression model to score event credibility from **312** hand labels, using a frozen hash-based test split; reached **0.68 ROC-AUC** (95% CI 0.57-0.77).
- Wrote **158 automated tests** run in **GitHub Actions CI** against PostgreSQL, with Docker Compose for local deployment.

Heart Disease Prediction System | *Python, scikit-learn, XGBoost, PostgreSQL, Streamlit, Docker* Sep 2026

- Built an end-to-end **machine learning pipeline** that predicts heart disease from 13 clinical features across **303 patient records**, reaching **96% recall** (27 of 28 caught) and **0.96 ROC-AUC** on held-out test data.
- Compared logistic regression, random forest, and XGBoost with **5x5 repeated cross-validation**, shipping logistic regression on a pre-set rule (**0.705 precision** at 90%+ recall vs. ~0.65 for tree models).
- Chose the decision threshold out of fold for a **90%+ recall target**, with **1,000-sample bootstrap** confidence intervals.
- Stress-tested the frozen model on **617 patients from 3 other hospitals**, where pooled recall fell to **0.70**.
- Deployed a public **Streamlit app** on a Dockerized **PostgreSQL** backend that logs model runs, with **22 automated tests**.

Personal Portfolio Website | *Next.js, React, TypeScript, Tailwind CSS* Oct 2026

- Built and deployed a responsive portfolio on Vercel using Next.js static export and MDX-driven content.
- Implemented content scrolling, section navigation, light/dark themes, and cursor lighting with reduced-motion support.
- Automated type checking, linting, content validation, tests, and link checking through GitHub Actions.

Information Systems Strategy & Database Project | *Team Lead, four-person team* Spring 2026

- **Connected four business workflows**, customer service, order processing, inventory, and logistics, by leading a four-person team in developing a **cloud-based ERP and CRM recommendation**.
- **Produced an implementation-ready systems plan** that turned a focus-differentiation strategy into a roadmap.

- **Modeled six core business entities and one end-to-end workflow** by creating an **ERD** for customers, orders, products, shipments, service tickets, and employees and mapping order fulfillment using BPMN.
- **Standardized analysis across team deliverables** by writing and testing SQL queries to prioritize service tickets, analyze purchase history, track orders in process, and calculate delivered revenue.
- **Optimized SQL queries** written by team members, leading to a **25%** reduction in execution time.